# JUNWOO PARK

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## EDUCATION

**Sungkyunkwan University (SKKU)** **Suwon, Korea**
*B.S. in Computer Science and Engineering*, GPA: 4.31/4.50 Mar. 2024 – Feb. 2026

**Sejong University** **Seoul, Korea**
*Aerospace Engineering* Mar. 2020 – Feb. 2024

- Fulfilled mandatory alternative military service (2022–2023) before transferring to SKKU.

## RESEARCH EXPERIENCE

**SKKU AI-Healthcare Lab** **Suwon, Korea**
*Research Assistant (Advisor: Prof. Sujee Lee)* Feb. 2025 – Present

- Developed a multimodal deep learning model that integrates molecular graphs, sequences, images, and spectra using attention mechanisms for toxicity prediction [1].
- Collaborated with Samsung C-Lab on the "Muscle Tracker" project.

**SKKU Human Language Intelligence Lab** **Suwon, Korea**
*Research Assistant (Advisor: Prof. JinYeong Bak)* Sep. 2024 – Dec. 2024

- Analyzed and preprocessed user activity data including diary entries, therapeutic game logs, and media consumption to characterize depression-related behavioral patterns.

## PUBLICATIONS

[1] **J. Park** and S. Lee, MoltiTox: a multimodal fusion model for molecular toxicity prediction, Frontiers in Toxicology, 2025.

## PROJECTS

**Project SKKU Character: AI Platform for University Mascot, SKKU**
*AI Engineer* Oct. 2025 – Present

- Secured 5M KRW (∼3,600 USD) in funding through the university entrepreneurship program.
- Developing an AI image generation system by fine-tuning a FLUX model on mascot datasets and implementing a FastAPI backend for generation requests.

**Muscle Tracker: AI-driven Analysis for Strength Training, SKKU**
*Research Assistant* Aug. 2025 – Nov. 2025

- Collected and preprocessed EMG and IMU sensor data from 14 subjects through a Python pipeline applying Butterworth filtering, interpolation, and RMS computation.
- Designed an algorithm to predict maximum muscle activation via sigmoid curve fitting and co-developed an exercise classification model with 96% accuracy using a temporal convolutional network.

**NotiSKKU: Student Announcement App, SKKU**
*Team Leader* Aug. 2024 – Nov. 2025

- Led a four-member team to develop a mobile app that aggregates campus announcements, based on a survey of over 60 students.
- Built the frontend using Flutter and launched the service on Google Play.

**VisualVroom: Alert System for Hearing-Impaired Drivers, SKKU**
*Team Leader* Apr. 2025 – Aug. 2025

- Directed a team to create a real-time auditory alert system for hearing-impaired drivers.
- Developed a Vision Transformer-based model to classify sound sources and their directions from smartphone audio input, achieving 83.9% accuracy.
- Integrated Google Speech-to-Text API to support in-car communication.

## TEACHING & MENTORING EXPERIENCE

**Course: Basis and Practice in Programming, SKKU**
*Teaching Assistant* Fall 2025 – Present

- Guided 60 students in C programming coursework, evaluating assignments and providing feedback.

**Software Mentoring Program, SKKU**
*Mentor* May – Sep. 2024, May – Sep. 2025

- Taught Python fundamentals and conducted lab sessions for high school students.

## EXTRACURRICULAR ACTIVITIES

**Google Developer Groups on Campus, SKKU**
*President* Feb. 2025 – Feb. 2026

- Led ∼20 members, fostering collaboration through workshops and team-based projects.
- Established an international partnership with Mission San Jose High School in California, organizing programming competitions in Winter 2025 and a joint hackathon planned for Spring 2026.
- Hosted monthly tech talks featuring professionals and researchers from global tech companies.

## AWARDS & HONORS

**First Place, 2025 Uni-DTHON (Hackathon)** Nov. 2025

**Excellence Award, SW High School Creative Science-thon** Nov. 2025

- Awarded for leading a team of 4 high school students as a mentor.

**2025 President's List, Sungkyunkwan University** Sep. 2025

**Grand Prize, ICT Award Korea 2025** Aug. 2025

## SKILLS

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|---|---|
| **Languages** | Python, C, C++, Java, SQL, HTML/CSS |
| **Frameworks & Tools** | PyTorch, FastAPI, Flutter, Git, Docker, Linux, MATLAB |